

Macroeconomics III
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Chapter 6

EDUCATION AND
GROWTH: THE
SOLOW MODEL WITH
HUMAN CAPITAL

INTRODUCTION

1. The steady state predictions of the Solow model seem quite supported by the data, even though...
 2. ...the investment ratio s appears in the data to have a very strong and positive impact on percapita GDP y , while population growth rate n has a very strong negative effect on y .
 3. The empirical results do not contradict convergence, but in reality **convergence is much slower** than predicted by the Solow model.
- Is there a way to modify the Solow model that can work to rectify **both** empirical problems?
 - Yes: according to Mankiw, Romer, Weil, 1992, 'A Contribution to the Empirics of Economic Growth', *QJE*, (MRW): considering **Human Capital**

- **Human capital:** the accumulated stock of what is invested in training of the labour force: education, learning by doing, health, etc.
- Assume that human capital is **accumulated in the same way as physical capital:** every year a constant share of GDP is used for investment in human capital. Then, accumulation of human capital should **affect the rate of convergence** in the same way as accumulation of physical capital.

- Human capital **cannot be separated from the workers**. One can separate a man from his computer but not from his education. Hence, **the return on human capital accrues to the workers**. This keeps the **"labour's share" unchanged** even though a new kind of capital is being accumulated.
- It looks as if human capital could help solve the problem concerning the rate of convergence in the Solow model.

- What about the problem concerning the **steady state prediction** of the Solow model?
- Consider an **increase in the rate of investment in physical capital** (called s until now). This increases capital and income per worker in steady state through the usual well-known channels.
- With a constant rate of investment also in human capital, **larger income per worker means more investment in human capital** per worker and, in the end, more human capital per worker in steady state. Since human capital is productive, **income per worker will increase more** than without human capital.

- Seemingly, human capital could also help solve the problem concerning the impact of the investment rate on GDP per capita
- Hence, incorporating human capital might solve **both** of the empirical problems of the Solow model.
- Furthermore, it is **an aim in itself** to improve the model by adding human capital, since, by intuition, **the skills of the labour force** must be an **important input** factor.

THE SOLOW MODEL WITH HUMAN CAPITAL

- More or less the same micro world as in the Solow model
- The same types of agents: one type of (representative) firms and one type of (representative) consumers - and possibly a government sector
- But **the firm now uses human capital in its production** and
- **the consumer also accumulates human capital.**
- Output is used for consumption, investment in physical capital **or investment in human capital.** There is still just one production sector.
- The same markets: output and (services of) physical capital and labour

- No separate market for human capital services since human capital is linked to labour and its **services are sold inseparably together with the labour**
- The services traded in the labour market are no longer man-years of "raw labour", but man-years endowed with human capital. The **real wage is a mix of compensation for raw labour and return to human capital**.
- The number of workers, the labour force, is L_t . Every worker is endowed with the same amount of human capital, h_t . The total level of human capital in the economy is $H_t = h_t L_t$.
- Raw labour input, L_t^d , cannot be varied independently of human capital input. The input of human capital, $h_t L_t^d$, is proportional to L_t^d because each worker brings her h_t . Here is an important distinction from physical capital.
- We can now describe the new elements of the economy.

THE PRODUCTION FUNCTION WITH HUMAN CAPITAL

$$Y_t = K_t^\alpha H_t^\varphi (A_t L_t)^{1-\alpha-\varphi} = K_t^\alpha (h_t L_t)^\varphi (A_t L_t)^{1-\alpha-\varphi} = K_t^\alpha h_t^\varphi A_t^{1-\alpha-\varphi} L_t^{1-\alpha},$$

where $0 < \alpha < 1$, $0 < \varphi < 1$, $\alpha + \varphi < 1$, $A_t = (1 + g)^t A_0$

- Constant returns to K_t, H_t, L_t : the **replication argument**
- The per capita production function is:

$$y_t = k_t^\alpha h_t^\varphi A_t^{1-\alpha-\varphi} \Rightarrow g_t^y = \alpha g_t^k + \varphi g_t^h + (1 - \alpha - \varphi) g_t^A.$$

- When calculating the marginal product of labour, one has to take into account that one additional worker comes with a given amount of human capital, h_t . Hence, h_t , not H_t , should be taken as given when we differentiate wrt. L_t .

- Remembering that $Y_t = K_t^\alpha h_t^\varphi A_t^{1-\alpha-\varphi} L_t^{1-\alpha}$ the returns to capital and labour, respectively, are computed as:

$$r_t = \alpha K_t^{\alpha-1} h_t^\varphi A_t^{1-\alpha-\varphi} L_t^{1-\alpha} = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} \left(\frac{H_t}{A_t L_t} \right)^\varphi$$

$$w_t = (1-\alpha) K_t^\alpha h_t^\varphi A_t^{1-\alpha-\varphi} L_t^{-\alpha} = (1-\alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{H_t}{A_t L_t} \right)^\varphi A_t.$$

- Observe that $r_t K_t / Y_t = \alpha$ and $\frac{w_t L_t}{Y_t} = 1 - \alpha$: the workers get the return to human capital.
- As usual, empirical observations suggest $\alpha \cong 1/3$, but also (a little less firm perhaps) $\varphi \cong 1/3$, or a bit larger: average and minimum wage rates in US manufacturing.

THE CONSUMERS' ACCUMULATION OF CAPITAL

- As before, each of the L_t consumers supplies one unit of labour inelastically and
- the number of consumers grows at a constant rate, $n > -1$,
- and all physical capital, K_t , is supplied as long as $r_t > 0$.
- As in the general Solow model, the representative consumer has to decide C_t given Y_t , which in turn determines $S_t = Y_t - C_t$.
- But now the consumer also has to decide how to divide gross savings, S_t , into gross investment in physical capital, I_t^K , and gross investment in human capital, I_t^H .

- Given I_t^K and I_t^H :

$$K_{t+1} - K_t = I_t^K - \delta K_t$$

$$H_{t+1} - H_t = I_t^H - \delta H_t$$

where we have assumed the same rate of depreciation, $0 < \delta < 1$, for physical and human capital.

- The restriction that I_t^K and I_t^H have to fulfill is:

$$I_t^K + I_t^H = Y_t - C_t = S_t$$

- We assume that the consumer's considerations result in the standard Solow assumptions:

$$I_t^K = s_K Y_t \quad \text{and} \quad I_t^H = s_H Y_t$$

$$0 < s_K < 1, \quad 0 < s_H < 1, \quad s_K + s_H < 1$$

- Hence $S_t = (s_K + s_H) Y_t \Rightarrow$

$$C_t = (1 - s_K - s_H) Y_t.$$

THE COMPLETE SOLOW MODEL WITH HUMAN CAPITAL

$$Y_t = K_t^\alpha H_t^\varphi (A_t L_t)^{1-\alpha-\varphi}$$

$$r_t = \alpha \left(\frac{K_t}{A_t L_t} \right)^{\alpha-1} \left(\frac{H_t}{A_t L_t} \right)^\varphi$$

$$w_t = (1-\alpha) \left(\frac{K_t}{A_t L_t} \right)^\alpha \left(\frac{H_t}{A_t L_t} \right)^\varphi A_t$$

$$K_{t+1} - K_t = I_t^K - \delta K_t, \quad K_0 \text{ given}$$

$$H_{t+1} - H_t = I_t^H - \delta H_t, \quad H_0 \text{ given}$$

$$L_{t+1} = (1+n)L_t, \quad L_0 \text{ given}$$

$$A_{t+1} = (1+g)A_t, \quad A_0 \text{ given}$$

- **Parameters:** $\alpha, \varphi, s_K, s_H, \delta, n, g$.
- **Given** K_0, H_0, L_0, A_0 **the model determines the sequences** $(K_t), (H_t), (L_t), (A_t), (Y_t), (r_t), (w_t), (S_t)$.

THE LAW OF MOTION

1. Define:

$$\tilde{k}_t \equiv \frac{k_t}{A_t} = \frac{K_t}{A_t L_t}, \quad \tilde{h}_t \equiv \frac{h_t}{A_t} = \frac{H_t}{A_t L_t}, \quad \tilde{y}_t \equiv \frac{y_t}{A_t} = \frac{Y_t}{A_t L_t}.$$

2. From $Y_t = K_t^\alpha H_t^\varphi (A_t L_t)^{1-\alpha-\varphi}$ we get that

$$\tilde{y}_t = \tilde{k}_t^\alpha \tilde{h}_t^\varphi.$$

3. Restatement of the capital accumulation equations:

$$K_{t+1} - K_t = I_t^K - \delta K_t$$

$$H_{t+1} - H_t = I_t^H - \delta H_t$$

4. Dividing on both sides by $A_{t+1} L_{t+1}$ gives:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s_K \tilde{y}_t + (1-\delta) \tilde{k}_t \right)$$

$$\tilde{h}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s_H \tilde{y}_t + (1-\delta) \tilde{h}_t \right).$$

5. Inserting $y_t = \tilde{k}_t^\alpha \tilde{h}_t^\varphi$ gives the **transition equations**:

$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s_K \tilde{k}_t^\alpha \tilde{h}_t^\varphi + (1-\delta) \tilde{k}_t \right)$$

$$\tilde{h}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s_H \tilde{k}_t^\alpha \tilde{h}_t^\varphi + (1-\delta) \tilde{h}_t \right).$$

- The law of motion: two coupled, first-order difference equations in $\tilde{k}_t$ and $\tilde{h}_t$. Given $\tilde{k}_0$ and $\tilde{h}_0$ they determine $(\tilde{k}_t)$ and $(\tilde{h}_t)$.
- There is no easy diagrammatic way to show convergence towards a steady state. We will show:
 1. There is a well-defined steady state.
 2. Numerical simulations suggest convergence towards a steady state for reasonable parameter values.
 3. Convergence holds for a linear approximation around a steady state.

6. Subtracting $\tilde{k}_t$ and $\tilde{h}_t$, respectively, on both sides of the transition equations gives the **Solow equations**:

$$\tilde{k}_{t+1} - \tilde{k}_t = \frac{1}{(1+n)(1+g)} \left(s_K \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n+g+\delta+ng) \tilde{k}_t \right)$$

$$\tilde{h}_{t+1} - \tilde{h}_t = \frac{1}{(1+n)(1+g)} \left(s_H \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n+g+\delta+ng) \tilde{h}_t \right).$$

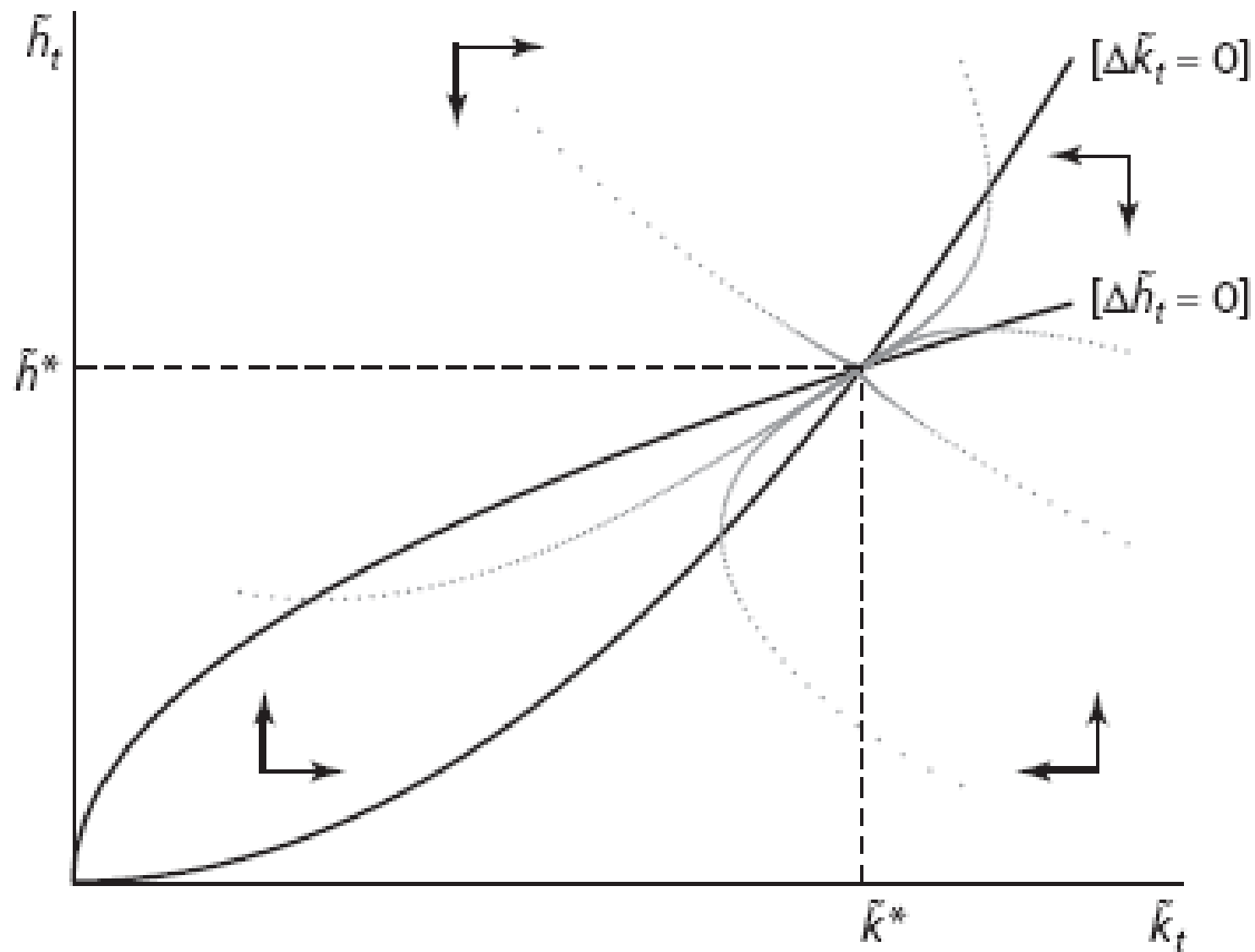
- From these we found in the previous lecture a well-defined steady state: $\tilde{k}^*$, $\tilde{h}^*$, implying also a steady state growth path, y_t^* , for income per worker.
- From the Solow equations also follow:

$$\Delta \tilde{k}_t = 0 \Leftrightarrow \tilde{h}_t = \left(\frac{n+g+\delta+ng}{s_K} \right)^{\frac{1}{\varphi}} \tilde{k}_t^{\frac{1-\alpha}{\varphi}},$$

$$\Delta \tilde{h}_t = 0 \Leftrightarrow \tilde{h}_t = \left(\frac{s_H}{n+g+\delta+ng} \right)^{\frac{1}{1-\varphi}} \tilde{k}_t^{\frac{\alpha}{1-\varphi}}$$

- We use these to construct:

The Phase diagram for the Solow model with human capital



STEADY STATE

- In steady state, $\tilde{k}_{t+1} - \tilde{k}_t = \tilde{h}_{t+1} - \tilde{h}_t = 0$:

$$s_K \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n + g + \delta + ng) \tilde{k}_t = 0$$

$$s_H \tilde{k}_t^\alpha \tilde{h}_t^\varphi - (n + g + \delta + ng) \tilde{h}_t = 0.$$

- Solving for $\tilde{k}_t = \tilde{k}^*$ and $\tilde{h}_t = \tilde{h}^*$ gives the steady state values:

$$\tilde{k}^* = \left(\frac{s_K^{1-\varphi} s_H^\varphi}{n + g + \delta + ng} \right)^{\frac{1}{1-\alpha-\varphi}}, \quad \tilde{h}^* = \left(\frac{s_K^\alpha s_H^{1-\alpha}}{n + g + \delta + ng} \right)^{\frac{1}{1-\alpha-\varphi}}.$$

- Inserting these into $y_t = \tilde{k}_t^\alpha \tilde{h}_t^\varphi$ gives:

$$\tilde{y}^* = \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{1-\alpha-\varphi}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\varphi}{1-\alpha-\varphi}}$$

and then, from $y_t = \tilde{y}_t A_t$, the steady state growth path:

$$y_t^* = A_t \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{1-\alpha-\varphi}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\varphi}{1-\alpha-\varphi}} .$$

- Taking logs gives:

$$\begin{aligned} \ln y_t^* &= \ln A_t + \frac{\alpha}{1-\alpha-\varphi} \left(\ln s_K - (n + g + \delta + ng) \right) \\ &\quad + \frac{\varphi}{1-\alpha-\varphi} \left(\ln s_H - (n + g + \delta + ng) \right) . \end{aligned}$$

- The elasticity of y_t^* with respect to:
 - s_K is now $\alpha / (1-\alpha-\varphi)$, where in the Solow model it was $\alpha / (1-\alpha)$. It is now around 1, where before it was around 1/2.
 - s_H is $\varphi / (1-\alpha-\varphi)$, that is, around one.
 - $(n + g + \delta)$ is now $-(\alpha + \varphi) / (1-\alpha-\varphi)$, where in the Solow model it was $-\alpha / (1-\alpha)$, that is, now around -2, where before it was around -1/2.

These features are much more realistic.

STRUCTURAL POLICIES FOR STEADY STATE

$$y_t^* = A_t \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{1-\alpha-\varphi}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\varphi}{1-\alpha-\varphi}} .$$
$$c_t^* = A_t (1 - s_K - s_H) \left(\frac{s_K}{n + g + \delta + ng} \right)^{\frac{\alpha}{1-\alpha-\varphi}} \left(\frac{s_H}{n + g + \delta + ng} \right)^{\frac{\varphi}{1-\alpha-\varphi}} .$$

1. The effects of s_K and $n + g + \delta$ on y_t^* are **qualitatively** as in the Solow model, but the **quantitative** effects are now much stronger and much more plausible. And $n + g + \delta$ has a stronger effect than s_K !
2. The effect of s_H is of the same size as the effect of s_K .
Golden rule:

$$s_K = \alpha \quad \text{and} \quad s_H = \varphi.$$

An interesting new question is:

3. Should the government promote investment in education (up to golden rule) by subsidizing education?
- Note: this happens to a large extent in many Western countries.
 - Also note: the empirics suggest that investment in human capital is (at least) as important as investment in physical capital.
 - Does this make investment subsidies in physical and human capital equally-well or equally-badly motivated?
 - A general view of welfare economics: intervene if (and only if) private decisions cannot be expected to lead to socially optimal outcomes.
 - Government subsidies have to be motivated by specific market imperfections.

- Some imperfections that specifically point to education subsidies:
 - **Imperfect credit markets** (adverse selection problem)
 - Imperfect insurance markets (moral hazard problem)
 - Externalities
 - Imperfect private knowledge
- Another motive for education subsidies lies in distributive concerns.
- If inequality is high, **the poor may not invest in education**. Underinvestment by the many poor cannot be compensated by over-investment by the few rich.
- Decreasing returns to education: can one rich study 100 years? Can this ever compensate 5 poor not studying 20 years each?

STABILITY OF STEADY STATE

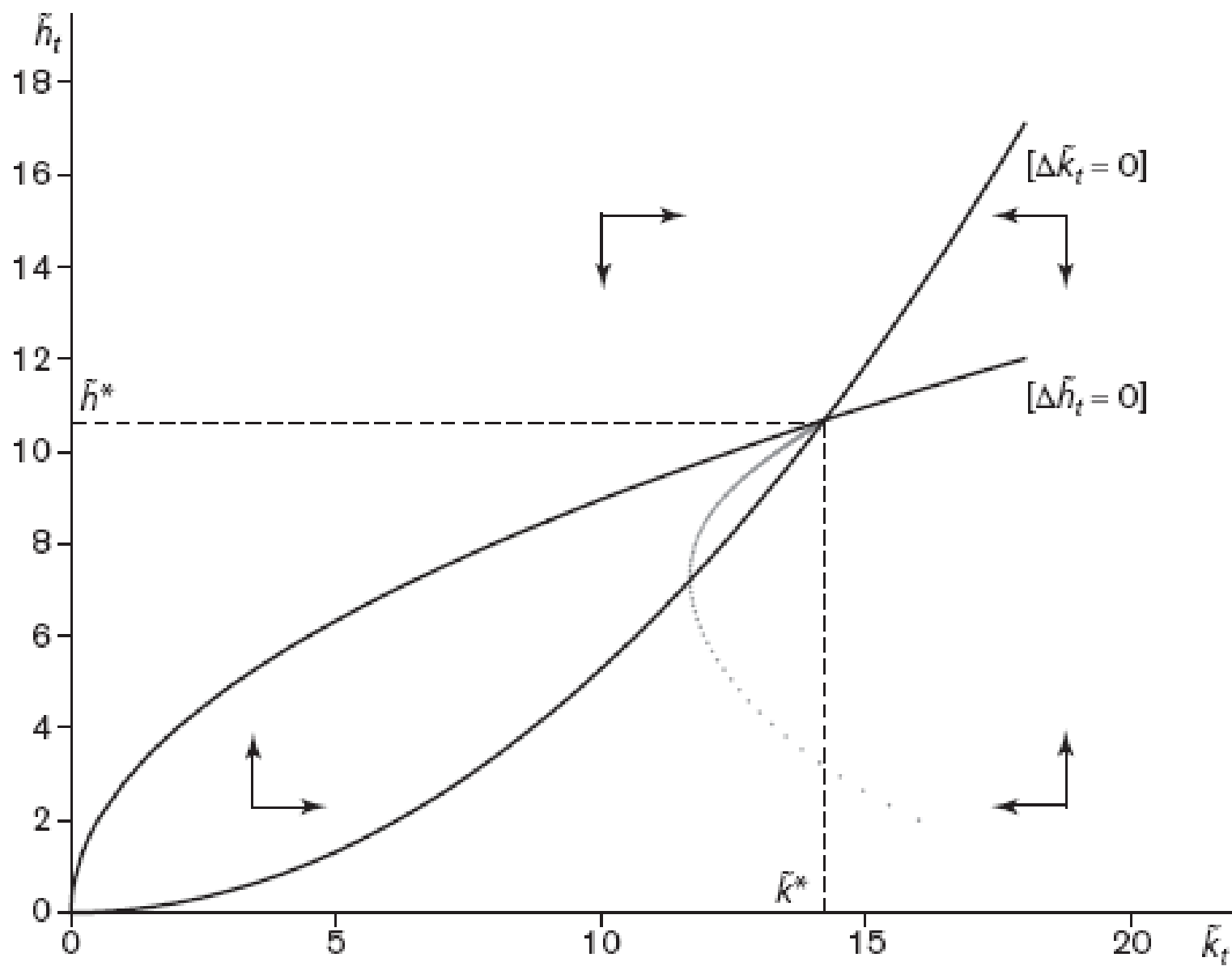
- In the previous part we saw that the steady state prediction does well empirically and has interesting policy implications. It also exhibits **balanced growth**:
 - $\tilde{k}_t, \tilde{h}_t$ and $\tilde{y}_t$ are constant, hence k_t , h_t , and y_t grow at a constant rate, g .
 - $w_t = (1 - \alpha) \tilde{k}_t^\alpha \tilde{h}_t^\varphi A_t \Rightarrow w_t$ grows at the constant rate, g , too.
 - $r_t = \alpha \tilde{k}_t^{\alpha-1} \tilde{h}_t^\varphi A_t \Rightarrow r_t$ constant.
- We don't know yet if the model implies convergence to steady state. We will show convergence in a **numerical** simulation

SIMULATION

- Let $\alpha = \varphi = 1/3$, $s_K = 0.2$, $s_H = 0.15$, $\delta = 0.06$, $n = 0$ and $g = 0.015$.
- In steady state: $\tilde{k}^* = 14.22$ and $\tilde{h}^* = 10.67$.
- Start in $\tilde{k}_0 = 16$ and $\tilde{h}_0 = 2$, and simulate using the transition equations:

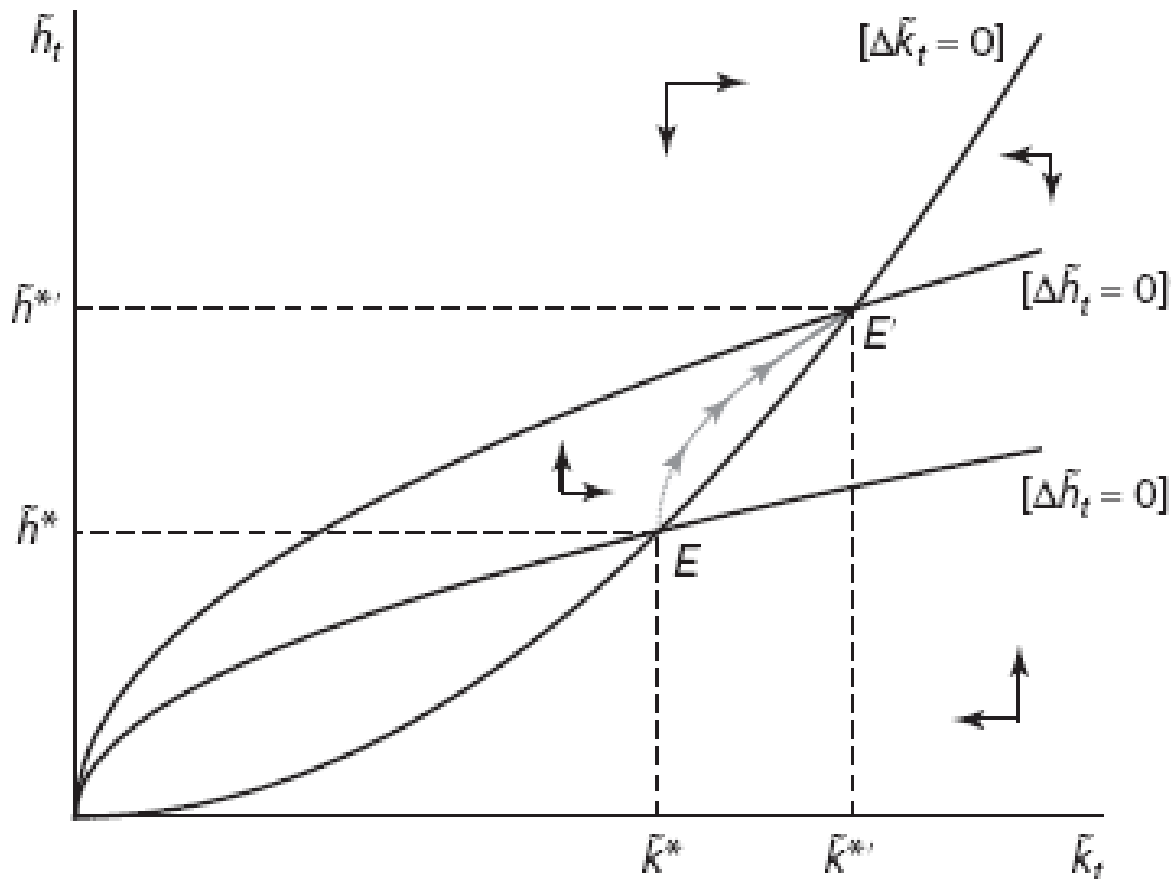
$$\tilde{k}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s_K \tilde{k}_t^\alpha \tilde{h}_t^\varphi + (1-\delta) \tilde{k}_t \right)$$
$$\tilde{h}_{t+1} = \frac{1}{(1+n)(1+g)} \left(s_H \tilde{k}_t^\alpha \tilde{h}_t^\varphi + (1-\delta) \tilde{h}_t \right).$$

- The result is shown in the phase diagram:



Comparative analysis in the phase diagram

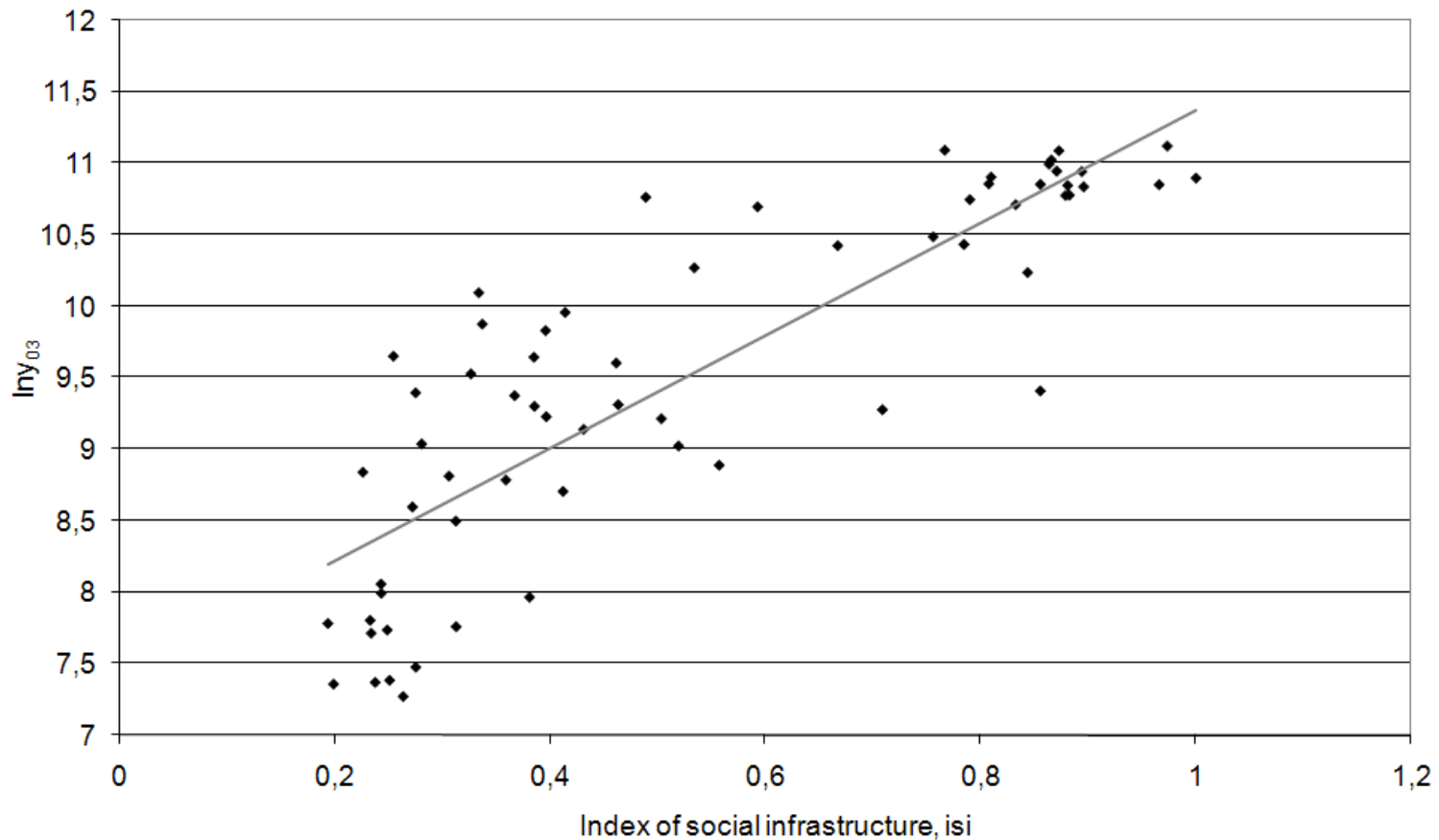
- An increase in s_H shifts $[\Delta \tilde{h}_t = 0]$ upwards. At first, only $\tilde{h}_t$ increases, but then $\tilde{k}_t$ begins to increase as well.



- What happens to $\tilde{y}_t$ and to $y = \tilde{y}_t A_t$? The growth rates jump!

SOCIAL INFRASTRUCTURE

- Why do some countries invest so much more than others?
And why are some countries much more productive?
- An explanation could be found (partly) in differences in social infrastructure or institutions:
 - Presence and quality of a financial system
 - Quality of government rules and regulations
 - Well-functioning systems of laws and courts
 - Sound fiscal policy with an uncorrupted expenditure side
 - Good educational system
- Plotting a constructed index of social infrastructure against GDP per worker unveils a clear positive correlation:



- Correcting for reversed causality results in an even stronger effect of institutions on economic performance.
- This suggests that institutions certainly are important when considering structural policy.

SUMMING UP

With the Solow model with human capital we have:

- Explained intuitively why incorporating human capital into the Solow model should work to overcome both of its empirical problems
- Set up explicitly a Solow model with accumulation of both physical and human capital
- Derived the model's law of motion
- Found its steady state and realised that indeed it conforms with our intuitions
- Tested the model's steady state prediction empirically and found it to perform remarkably well

- Discussed the model's implications for structural policy:
 - Traditional policies for saving and investment in physical capital and for population growth are qualitatively supported as in the Solow model, but with stronger and more plausible qualitative effects.
 - Policies for investment in human capital are supported as strongly as policies for investment in physical capital. Given the imperfections related to educational decisions, public subsidies to education seem to be well-motivated.

CONCLUSIONS

1. The Solow model with human capital performs very well empirically, both wrt. its steady state prediction and wrt. its convergence prediction, even under the assumption of "same technology" in all countries.
2. This is reassuring for policy implications!
3. However, important parameters are unexplained, e.g., s_K and s_H .
4. The quality of "institutions" may provide explanation of otherwise unexplained parameters.
5. The rate of technological growth that determines the long-run rate of growth in GDP per worker, is unexplained in the Solow model with human capital: "we have understood almost everything, and yet almost nothing!".
6. Points towards theories of endogenous technological progress.